

Diagonal Isoduality and Transversal Clifford Groups of MDS–CSS Codes

Tavis Rudd

August 2026

Abstract

Let C be a linear $[2m, m, m+1]_q$ maximum-distance-separable code. The space of diagonal multipliers carrying C to $C^\perp$ has dimension zero or one, and every nonzero multiplier has full support and is unique up to scalar. Over an odd prime field this nullity test determines the exact fixed-party projective transversal logical group of the associated $[[2m-1, 1, m]]_q$ MDS–CSS code: it is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ in the diagonally isodual case and $\mathbb{F}_q^2 \rtimes T$ otherwise, where T is the split torus. Rigidity of stabilizer absolutely maximally entangled states supplies the converse: every tensor-product logical implementation is Clifford factor by factor. The linear $\mathrm{SL}_2(q)$ action has coherent Weil lifts, while the full affine group retains the Heisenberg obstruction to a scalar splitting.

For six coordinates, diagonal isoduality is exactly self-association of the corresponding six-arc and hence the conic boundary. On an explicit non-GRS pencil a degree-eight quotient z classifies projective and monomial-code equivalence over odd fields and local-Clifford and local-unitary equivalence over odd prime fields. The Clebsch $[6, 3, 4]_{11}$ code gives a worked syndrome-geometric application. Fixed-copy scalar contractions are generically constant on the pencil, explaining why the operator-valued Weyl atlas retains classification data that bounded scalar invariants lose. Exact certificate replays cover the finite six-point computations; the all-length multiplier and group theorems are conceptual.

Index Terms—Absolutely maximally entangled states, Calderbank–Shor–Steane codes, conics, isodual codes, quantum MDS codes, self-association, six-arcs, transversal logical gates.

1 Introduction

A linear $[2m, m, m+1]_q$ maximum-distance-separable (MDS) code $C \leq \mathbb{F}_q^{2m}$ defines an equal-phase Calderbank–Shor–Steane (CSS) state

$$|\Psi_C\rangle = q^{-m/2} \sum_{c \in C} |c\rangle. \quad (1.1)$$

It is an absolutely maximally entangled state, and choosing one tensor factor as an input turns it into the normalized Choi state of a $[[2m-1, 1, m]]_q$ quantum MDS encoder. We determine the logical unitaries implemented by products of one-site physical operators.

Throughout, *transversal* means a tensor product of one-site physical operators with party labels fixed. Logical groups are projective, so scalar phases are divided out. A party-moving symmetry additionally permutes tensor factors and is named explicitly. The exact group has two branches, detected by a linear nullity computation.

Theorem 1.1 (Diagonal isoduality and the transversal group). Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. Put

$$\mathcal{D}(C, C^\perp) = \{s \in \mathbb{F}_q^{2m} : \mathrm{diag}(s)C \subseteq C^\perp\}.$$

This space has dimension zero or one. Its dimension is one exactly when C is diagonally isodual, meaning that $SC = C^\perp$ for a nonsingular diagonal matrix S . The projective group of logical unitaries admitting transversal physical implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \dim \mathcal{D}(C, C^\perp) = 1, \\ \mathbb{F}_q^2 \rtimes T, & \dim \mathcal{D}(C, C^\perp) = 0, \end{cases} \quad T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}.$$

Every nonzero diagonal-duality witness has full support and is unique up to scalar. On the isodual branch the linear $\mathrm{SL}_2(q)$ factor has a coherent Weil lift, whereas the affine projective group has no homomorphic lift through the scalar quotient because of the Weyl commutator.

The proof separates existence from exclusion. A nonzero off-diagonal block of a product Clifford produces a diagonal multiplier between C and $C^\perp$. The MDS distance forces that multiplier to have full support and forces the multiplier space to be a line. If the line is absent, every block is diagonal and the exact linear image is T . If it is present, the multiplier and its inverse construct all upper and lower elementary unipotents, hence all of $\mathrm{SL}_2(q)$. Logical Paulis supply the translation factor $\mathbb{F}_q^2$. The converse uses the following rigidity theorem from the companion paper.

Complementary work describes transversal Clifford structure through compatible symplectic bases for self-dual CSS codes [TTF25, Theorem 1], diagonal multi-block Clifford symmetries and endomorphism algebras of qubit stabilizer codes [DB25, Theorem 6.1], and code automorphisms or ZX-dualities with the required Pauli corrections [SKW⁺25, Sections III–IV]. Albert gives normal forms for the full code-preserving, transversal, and two-fold-transversal Clifford groups of CSS codes [Alb26, Sections III–V]. For AME-derived codes, Tan computes the four-qutrit and six-qubit symmetry/transversal dictionaries in concrete cases [Tan26b, Section 5] and [Tan26a, Section 3.4]. These results concern qubit CSS/stabilizer structure or individual AME examples. Here the one-logical-qudit odd-prime MDS hypothesis reduces the full fixed-party product-unitary problem to the nullity of one diagonal code-to-dual multiplier space; stabilizer-AME rigidity then makes the resulting Clifford image exhaustive among arbitrary product unitaries.

Theorem 1.2 (Imported stabilizer-AME rigidity [Rud26a]). For every prime power q and $m \geq 2$, every product-unitary intertwiner between two stabilizer $\mathrm{AME}(2m, q)$ states is Clifford on each tensor factor.

Corollary 1.3 (Imported transversal Clifford no-go [Rud26a]). Every tensor-product conversion between the associated stabilizer $[[2m - 1, 1, m]]_q$ encoders is Clifford on every physical factor and on the logical factor. In particular, no code in this family admits a transversal non-Clifford logical unitary.

We also use the companion paper’s phase-correction lemma: if a product Clifford carries one stabilizer label group to another, a product Pauli correction carries the corresponding stabilizer state rays.

Lemma 1.4 (Imported stabilizer-character correction [Rud26a]). For stabilizer states in the standard finite-field Weyl convention, every product-Clifford map of their stabilizer label groups can be corrected by a product Pauli to map one state ray to the other.

Imported minimum-support atlas [Rud26a]. For a stabilizer $\mathrm{AME}(2m, q)$ state with label group L , every $(m + 1)$ -party subgroup $L(A)$ projects bijectively to the local Pauli-label plane at each party in A , and these subgroups span L . After a party relabelling, two such states are local-Clifford equivalent exactly when their transition maps $p_j p_i^{-1}$ are related by local trace-symplectic changes of Weyl frame.

We use only these stated conclusions. Their proofs, quantitative rounding theory, and minimum-support atlas belong to the companion paper and are not reproved here. The MDS–CSS dictionary and all constructions that distinguish the two exact Clifford branches are proved in this paper.

For generalized and extended generalized Reed–Solomon codes the standard dual formula supplies a diagonal-duality witness, so the first branch recovers the full one-qudit projective Clifford group. An extended $[8, 4, 5]_7$ Reed–Solomon code, for example, gives a $[[7, 1, 4]]_7$ quantum code whose projective transversal logical group has order $7^2|\mathrm{SL}_2(7)| = 16464$. The nullity criterion also applies to non-evaluation MDS codes.

At length six, the code has a projective-geometric model as a six-arc in $\mathbb{P}^2(\mathbb{F}_q)$. Diagonal isoduality becomes fixed-label Gale self-association, which for a six-arc is equivalent to lying on a conic. Thus the all-length group theorem specializes to the $\mathrm{SL}_2(q)$ -versus-split-torus logical phase. An explicit non-GRS pencil then admits a degree-eight quotient z : over odd fields it classifies projective and monomial-code equivalence, and over odd prime fields the imported rigidity theorem upgrades the classification to local Clifford and local unitary equivalence. The extension-field Frobenius sectors are stated separately; no full extension-field Clifford classification is claimed.

The remaining results explain and test this structure. The Clebsch $[6, 3, 4]_{11}$ code relates extremal generalized- X syndromes to twelve conic directions while remaining on the nonconic split-torus branch. Exact marginal and four-copy contractions distinguish selected scalar orbits, but every fixed-copy scalar contraction is generically constant on a regular code family. This contrast explains why operator-valued Weyl-frame data can classify where bounded scalar invariants cannot.

Section 2 fixes the MDS–CSS and six-arc dictionary. Section 3 proves the multiplier-line theorem and the exact group. Sections 4–6 develop the six-point applications and scalar boundary. The appendices contain the finite separators, transport calculation, and party-extension tables.

A Lean 4 development, entered through a single import-only gate, checks the dictionary, the multiplier line and nullity test, the syndrome geometry, the pencil quotient, and the extension-field scalar algebra, and states the remaining exact-carrier, classification, separator, and transport results as interfaces over hypotheses proved here or certified in the supplement. Section 7 names the gate, its axiom audit, and what each category leaves trusted.

2 Stabilizer AME states and MDS–CSS models

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

After choosing dual $\mathbb{F}_p$ -bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p -level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \mathrm{PCliff}(q) \longrightarrow \mathrm{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \quad (2.0)$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$ -linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n . Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \quad (2.1)$$

If A is a party set, put

$$L(A) = \{\ell \in L : \text{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \quad (2.2)$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

The rigidity and atlas theorems use only (2.0)–(2.2). For the geometric classification and explicit six-party applications, we now specialize to the MDS–CSS model.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H . The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \text{rowspan } H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

For the resulting CSS stabilizer we use the exact lift

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^6 X(c_i)Z(h_i), \quad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$.

Proposition 2.1 (Arc–AME dictionary). For a six-arc with kernel C , the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer $\text{AME}(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Conversely, for a linear $[6, 3]_q$ code C the state $|\Psi_C\rangle$ is $\text{AME}(6, q)$ only if C has minimum distance four, so the dictionary is an equivalence. Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I , puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is $\text{AME}(6, q)$. Its support has the minimum possible size q^3 .

For the converse, suppose the linear $[6, 3]_q$ code C has a nonzero word c of weight at most three, and choose a three-party set A containing its support. Then $(c, 0)$ is a nonidentity label of L_C supported on A , so $L_C(A) \neq \{0\}$ and ρ_A is not maximally mixed. Hence an AME equal-phase state forces minimum distance four, which at length six and dimension three is the exact $[6, 3, 4]_q$ parameter set.

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S , let

$$L_C(S) = \{\ell \in L_C : \text{supp}(\ell) \subseteq S\}.$$

The general stabilizer partial-trace formula specializes to

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \quad (2.3)$$

For later block calculations, write

$$C_X = C \oplus 0, \quad C_Z^\perp = 0 \oplus C^\perp, \quad L_C = C_X \oplus C_Z^\perp.$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use “GRS state” for the state of any such code. The non-GRS pencil in Section 4 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Exact MDS–CSS logical images and lifts

Proposition 3.1 (The diagonal-multiplier line). Let $E, F \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes, and put

$$\mathcal{D}(E, F) = \{s \in \mathbb{F}_q^{2m} : \text{diag}(s)E \subseteq F\}.$$

Every nonzero $s \in \mathcal{D}(E, F)$ has full coordinate support and $\text{diag}(s)E = F$. Consequently

$$\dim_{\mathbb{F}_q} \mathcal{D}(E, F) \leq 1, \quad \mathcal{D}(E, E) = \mathbb{F}_q(1, \dots, 1).$$

In particular, $\mathcal{D}(C, C^\perp)$ has dimension zero or one, and C is diagonally isodual exactly in the dimension-one case. When that line exists, any two nonzero witnesses differ by a unique nonzero scalar, so every ratio s_i/s_j is intrinsic.

Proof. Let $s \neq 0$, and write $D = \text{diag}(s)$. If D had support of size at most m , choose a word of E nonzero at a supported coordinate. Its image would be a nonzero word of F of weight at most m , contrary to the distance of F . Thus D has at least $m+1$ nonzero entries. Its kernel on E consists of words supported on at most $m-1$ coordinates, so the distance of E

makes that kernel zero. Equal dimensions now give $DE = F$. A zero diagonal entry would force every word of F to vanish at that coordinate, which an MDS code does not, so s has full support.

If two multipliers were independent, a nonzero linear combination could be chosen to vanish at one coordinate, contradicting full support. Hence $\dim \mathcal{D}(E, F) \leq 1$. The all-ones vector belongs to $\mathcal{D}(E, E)$, proving the self-multiplier statement. Taking $F = C^\perp$ gives the remaining assertions. $\square$

Proof of Theorem 1.1. Write a fixed-party product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, & D_\gamma C &\subseteq C^\perp, \\ D_\beta C^\perp &\subseteq C, & D_\delta C^\perp &\subseteq C^\perp. \end{aligned} \quad (3.1)$$

If the block at the chosen input is nondiagonal, then D_β or D_γ is nonzero. Proposition 3.1 and (3.1) give

$$D_\gamma C = C^\perp \quad \text{or} \quad D_\beta C^\perp = C.$$

The second equality can be inverted, so either one supplies a nonsingular diagonal S with $SC = C^\perp$. Consequently, on the non-isodual branch every input block is diagonal. Its determinant is one, and the common blocks $\text{diag}(a, a^{-1})$ always preserve L_C ; hence the exact linear factor there is T .

Now suppose

$$S = \text{diag}(s_1, \dots, s_{2m}), \quad SC = C^\perp. \quad (3.2)$$

The projective vector $[s_1 : \dots : s_{2m}]$, and hence every propagation ratio s_i/s_j , depends only on C .

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^-(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \quad F_i^+(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \quad (3.3)$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^\perp)$, the product of the lower blocks sends

$$(c, h) \mapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \mapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries: Lemma 1.4 supplies the required product-Pauli corrections. Thus label-space preservation is sufficient here; no coherent choice of Clifford phases is being assumed. At any chosen input party j , their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\text{SL}_2(q)$. Under the Choi correspondence, an input block A induces the logical block A^{-T} . This map is an automorphism of $\text{SL}_2(q)$, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \text{SL}_2(q)$ is transversal.

The imported Corollary 1.3 excludes every product-unitary logical action outside that group. Hence the displayed group is exact.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (3.2), including the extended evaluation point at length $q+1$ [HP03, Sections 5.2–5.3]. Thus the GRS tower lies on the diagonally isodual branch, but the proof used no evaluation structure.

There are two different lift questions behind this projective statement. Fix an input coordinate j , put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1}b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \quad (3.4)$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (3.3) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. This calculation also applies at the extended evaluation point. It uses only the diagonal duality $SC = C^\perp$, not the evaluation formula for a GRS code.

For the present odd prime q , and more generally for any odd prime power, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1]. Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A . In the symmetric Weyl convention $D_{a,b} = \chi(ab/2)X(a)Z(b)$, a CSS label $(c, h) \in C \oplus C^\perp$ has tensor lift

$$\bigotimes_i D_{c_i, h_i} = \chi(c \cdot h/2)X(c)Z(h) = X(c)Z(h),$$

because $c \cdot h = 0$. Exact Egorov covariance therefore preserves the CSS stabilizer group with its trivial character, and hence the state ray, rather than only its projective label space. Consequently the one-qudit Clifford scalar extension, restricted to the linear $\mathrm{SL}_2(q)$ factor, splits coherently with the propagated tensor symmetries. The exceptional uniqueness clause at $q = 3$ in Gérardin’s theorem does not remove existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \quad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent lift above is therefore $H_q \rtimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Neither conclusion decides the party-permutation extension. Its quotient is the evaluation-set-dependent realized permutation group Π , and its kernel is the full fixed-party projective group Γ , rather than the scalar phases. Its splitting remains exactly the nonabelian factor-set triviality criterion of Appendix C. $\square$

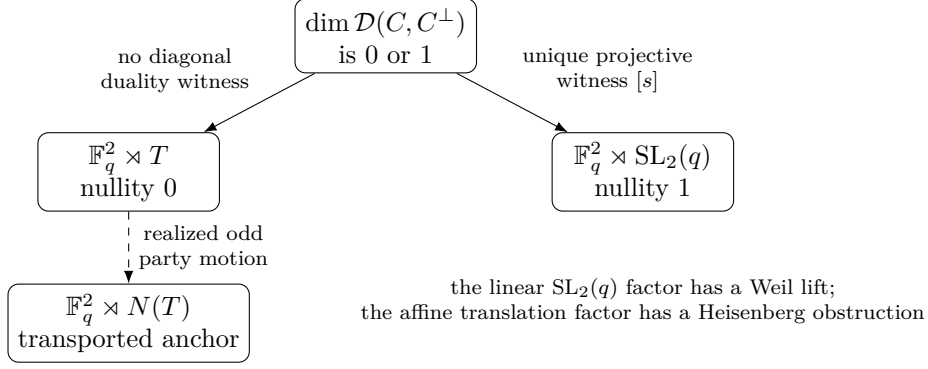


Figure 1: The nullity of the diagonal multiplier space selects the exact fixed-party projective transversal group. Party motion is a separate extension problem: in the computed non-GRS rows, an odd permutation enlarges the transported anchor carrier from T to $N(T)$, but it need not fix the chosen encoder input.

4 The non-GRS pencil and its quotient

Our first six-party application combines the minimum-support holonomy calculation with the imported rigidity theorem to turn a classical projective quotient into a complete LU coordinate over odd prime fields. Over a finite field of odd order, put

$$H_t = ((0, 1, 1 - t), (0, 1, t - 1), (1, 1 - t, 0), (1, t - 1, 0), (1, 0, -t), (1, 0, t)),$$

and define

$$A(t) = -4t(t - 1)^2, \quad B(t) = (t^2 - t + 1)(t^2 - 3t + 1), \quad z(t) = (B(t)/A(t))^2.$$

Also put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t - 1)(t^2 - t + 1)(t^2 - 3t + 1)G(t) \neq 0. \quad (4.1)$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G , so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, with A, B, z as above.

Theorem 4.1 (Projective and Clifford quotient). For admitted t, u , the following are equivalent:

$$H_t \sim_{\text{proj}} H_u, \quad C_t \sim_{\text{mon}} C_u, \quad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\text{LC}} \Psi_u$.

Corollary 4.2 (Local classification of the admitted pencil). If q is an odd prime and t, u are admitted, then

$$H_t \sim_{\text{proj}} H_u \iff C_t \sim_{\text{mon}} C_u \iff \Psi_t \sim_{\text{LC}} \Psi_u \iff \Psi_t \sim_{\text{LU}} \Psi_u \iff z(t) = z(u),$$

where every equivalence allows a party permutation.

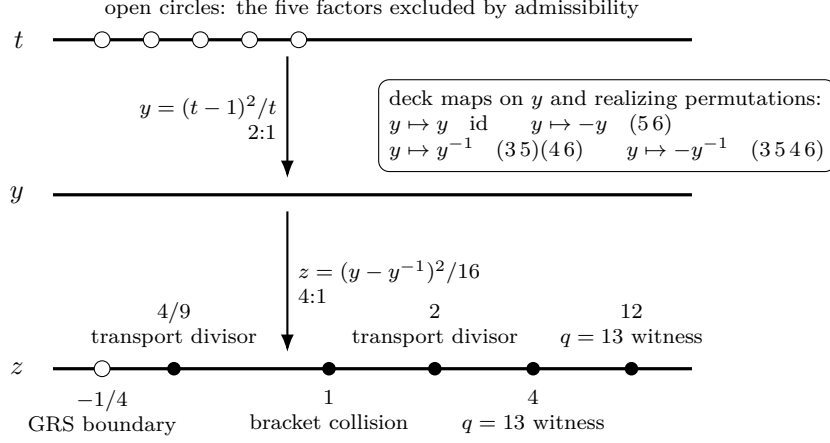


Figure 2: The degree-eight quotient of the admitted pencil (special points on the z -line are arranged schematically). The parameter t double-covers $y = (t - 1)^2/t$, and the four identifications $y \mapsto \pm y^{\pm 1}$, each realized by a displayed party permutation, cover z . The open point $z = -1/4$ is the excluded GRS boundary; $z = 1$ is a bracket collision; $z = 4, 12$ form the $q = 13$ four-copy witness pair; and $z = 2, 4/9$ belong only to the transport-rank detector, not to the LU-orbit boundary.

Proof. Set

$$y(t) = \frac{(t - 1)^2}{t}.$$

Then $A = -4t^2y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \quad (4.2)$$

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\text{diag}(1, (1 - u)/(1 - t), u/t)$
$y(u) = -y(t)$	$\text{diag}(1, (1 - u)/(1 - t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}.$

The corresponding permutations are respectively the identity, $(5\ 6)$, $(3\ 5)(4\ 6)$, and $(3\ 5\ 4\ 6)$. All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \quad (4.3)$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. On the admitted locus $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two in (4.3) distinguish the two opposite pairs and

recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset instead collapses to two opposite values, each of multiplicity five. That collapsed pattern is itself invariant under the common scaling and sign, and it is equivalent to $z = 1$. Thus the multiset recovers z in both cases, and projective equivalence forces equality of z . The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

We now specialize the imported minimum-support atlas to $m = 3$. Each four-party support identifies its two-dimensional stabilizer plane with the local Pauli-label plane at any party in that support. Overlapping supports therefore give coordinate changes between local Weyl frames, and their compositions around a loop give holonomy up to local symplectic conjugacy. These minimum-support planes generate the full stabilizer label group. The 450 quantities below are conjugacy data of this atlas, not an independent choice of scalar contractions.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_t}(T)$ for every four-set T . If $i, j \in T$, the imported atlas statement says that projection $p_i : K_T \rightarrow \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism. Define the transition

$$M_T(i, j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j , the holonomy

$$\mathrm{Hol}(T, U; i, j) = M_U(j, i) M_T(i, j). \quad (4.4)$$

Local symplectic blocks F_i send this endomorphism to $F_i \mathrm{Hol} F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Substituting the shortened generators into (4.4), simplifying each of the 450 rational functions exactly, and collecting equal functions gives the following multiset. The paper-local certificate described in Section 7 replays both this symbolic enumeration and a separate direct-Lagrangian reconstruction:

value of $r(M)$	multiplicity	(4.5)
4	90	
$-1/z$	144	
$(2z+1)^2/z^2$	24	
each root of $X^2 - 8X + (8 - 16z - 1/z)$	96.	

Because $A, B \neq 0$ on the admitted locus, $z \neq 0$. The two roots are taken over an algebraic closure; their symmetric multiset descends to $\mathbb{F}_q$. The multiplicity 96 applies to each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is 144, 168, 240, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z . $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (4.2), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w , and their projectivities lift to monomial local Cliffords.

Proof of Corollary 4.2. Theorem 4.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.2 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. $\square$

Proposition 4.3 (Frobenius-sector divisors). Let σ be an automorphism of a finite field of odd order and let t be admitted and non-GRS. Put

$$E_\sigma(t) = (\sigma(t) - t)(1 - \sigma(t)t), \quad D_\sigma(t) = ((1 - \sigma(t))(1 - t))^2 + \sigma(t)t.$$

Then

$$\frac{\sigma(t)}{(1 - \sigma(t))^2} = \frac{t}{(1 - t)^2} \iff E_\sigma(t) = 0. \quad (4.6)$$

If $h = (1 - \sigma(t))(1 - t)$ and $w = (-h, -h, 1, 1, -1, -1)$, then

$$H_{\sigma(t)} \text{diag}(w) H_t^\top = 0 \iff D_\sigma(t) = 0. \quad (4.7)$$

Moreover $D_{\text{id}}(t) = G(t)$, and

$$E_\sigma(t) = 0 \implies D_\sigma(t) \neq 0. \quad (4.8)$$

Finally G, A, B, z commute with σ , and σ preserves the admitted non-GRS locus.

Proof. Admission gives $t \neq 1$, and injectivity of σ gives $\sigma(t) \neq 1$. Cross multiplication in (4.6) gives

$$s(1 - t)^2 - t(1 - s)^2 = (s - t)(1 - st)$$

with $s = \sigma(t)$. Direct multiplication in (4.7) makes eight entries vanish identically and leaves $-2D_\sigma(t)$ in the last entry; odd characteristic gives the equivalence. Setting σ to the identity yields $D_{\text{id}} = G$.

If $E_\sigma(t) = 0$, then either $\sigma(t) = t$ or $\sigma(t)t = 1$. In the first case $D_\sigma(t) = G(t)$. In the second, multiplying $D_\sigma(t)$ by t^2 gives $G(t)$. Admission excludes $t = 0$ and $G(t) = 0$, proving (4.8). The equivariance statements follow by applying σ to the defining polynomial and rational expressions. $\square$

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p} , while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \quad z(-s) = 3 + 2s.$$

For each field automorphism σ , Proposition 4.3 gives the collision and diagonal-isodual divisors relating t to $\sigma(t)$. It does not show that an arbitrary additive local Clifford determines a nontrivial field automorphism, nor does it construct a local Clifford from a Galois match of z . Those two representation-theoretic bridges are not claimed here, so neither is a full extension-field LC/LU classification.

5 Diagonal isoduality and the logical-Clifford phase

The second application uses the atlas gauges to determine the exact fixed-party transversal logical group of an MDS-CSS encoder. Throughout this section q is an odd prime. Regard any one party of an $\text{AME}(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, the symplectic action of a

one-qudit Clifford is a block in $\text{SL}_2(q)$, so the kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \dots, F_6) \in \text{SL}_2(q)^6 : (\bigoplus_i F_i)L_C = L_C\}.$$

For $q = p^e$ with $e > 1$, the full local Clifford group acts instead through $\text{Sp}_{2e}(\mathbb{F}_p)$; the displayed field-linear block group is then not the full local Clifford kernel. For an encoder view, the *fixed-party logical symplectic image* means the image of $\mathcal{K}_C$ on its input block. Theorem 1.1 identifies the all-length phase boundary as diagonal isoduality. For six-arcs, classical self-association turns that intrinsic code condition into the conic criterion below.

Lemma 5.1 (Fixed-label self-association). Let H be a rank-three 3×6 matrix whose columns are a six-arc, and put $C = \ker H$. There is a nonsingular diagonal matrix S with $SC = C^\perp$ if and only if the six columns of H lie on a conic.

Proof. Let G be a 3×6 generator matrix for C . Its columns are the labeled Gale transform of the columns of H . Since $C^\perp = \text{row}(H)$,

$$SC = C^\perp \iff \text{row}(GS) = \text{row}(H) \iff AGS = H \text{ for some } A \in \text{GL}_3(q).$$

Thus the condition is precisely fixed-label projective equivalence to the Gale transform. This fixed locus can be identified directly over the finite field. Put

$$V_i = (x_i^2, y_i^2, z_i^2, x_i y_i, x_i z_i, y_i z_i)$$

and let V be the 6-by-6 matrix with rows V_i . Its determinant vanishes exactly when a nonzero quadratic form contains all six columns. Every five rows of V are independent: a relation supported on at most five points would give $H_I D H_I^T = 0$ for a nonsingular diagonal D on its support I ; for $|I| \geq 3$, this would put the three-dimensional row space of $H_I D$ inside $\ker H_I$, which has dimension at most two, while smaller supports are immediate. Hence any relation $\sum_i d_i V_i = 0$ has every $d_i \neq 0$.

Writing $D = \text{diag}(d_i)$, that relation is $HDH^T = 0$. Therefore $\text{row}(HD) \subseteq \ker H = C$, and equality follows from dimension three. Thus $CD^{-1} = C^\perp$. Conversely, a nonsingular diagonal equivalence $SC = C^\perp$ gives $HS^{-1}H^T = 0$, hence a relation among the rows of V and a conic through the six columns. The conic is irreducible because the columns form an arc. This is the finite-field fixed-label form of the classical six-point self-association theorem [HMSV08, Section 2.3]. $\square$

Theorem 5.2 (Fixed-party logical phase). For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party logical symplectic image is

$$\begin{cases} \text{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}, & \text{if the arc is nonconic.} \end{cases} \quad (5.1)$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-party symmetries. Off the GRS locus, the fixed-party logical symplectic image is the split torus. At a chosen anchor leg, any party-moving isoduality that exchanges the CSS X - and Z -spaces transports the encoder view by the nontrivial Weyl-group element, so the possible anchor blocks generate the normalizer of T together with T . This is an action on one fixed encoder only when the isoduality fixes its input leg.

Proof. Theorem 1.1 gives $\text{SL}_2(q)$ exactly on the diagonally isodual branch and T otherwise. Lemma 5.1 identifies diagonal isoduality with the conic locus. At a chosen anchor leg, a party-moving isoduality exchanges the CSS X - and Z -axes and hence represents the nontrivial Weyl-group element normalizing T . It acts on one fixed encoder only when the party motion fixes that encoder's input. $\square$

Adding logical Paulis recovers the exact fixed-party projective logical groups in Theorem 1.1.

This is an operational distinction, not a classical automorphism count. The fixed-party non-GRS group has order $q^2(q-1)$, while the GRS group has order $q^2|\mathrm{SL}_2(q)|$.

Lemma 5.3 (Coset states and three-party syndrome charts). Let H be a rank-three parity-check matrix whose six columns form an arc, put $C = \ker H$, and set

$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \quad |\Psi_C\rangle = q^{-3/2} \sum_{c \in C} |c\rangle.$$

Then $|\Psi_e\rangle = |\Psi_f\rangle$ if and only if $e - f \in C$, and states from distinct cosets are orthogonal. In the row basis of H , the $Z(C^\perp)$ -stabilizer syndrome of $|\Psi_e\rangle$ is $He^\top$. Consequently the least support of an X -operator producing $|\Psi_e\rangle$ is the minimum Hamming weight in $e + C$.

For every syndrome $s \in \mathbb{F}_q^3$ and every three-party set I , there is a unique representative e supported in I with $He^\top = s$. If the coset with syndrome s has minimum weight three, this representative has support exactly I .

Proof. Translation replaces the computational support C by the affine coset $e + C$. Two such supports are equal precisely when $e - f \in C$; otherwise they are disjoint, which also proves orthogonality. If h is a row of H , then $h \in C^\perp$, and $Z(h)$ acts on the entire coset $e + C$ by the character $\chi(h \cdot e)$. Reading this for the three rows of H gives $He^\top$.

Now fix I . The arc condition says exactly that the three columns in I form an invertible matrix H_I . Thus $H_I e_I^\top = s$ has one solution, and extension by zero gives the unique representative supported in I . If one of its three coordinates vanished, the same syndrome would have a representative of weight at most two. A minimum weight-three syndrome therefore uses all three coordinates. $\square$

Proposition 5.4 (Clebsch extremal X -syndrome conic). Let H be a parity-check matrix whose columns are the Clebsch six-arc in $\mathbb{P}^2(\mathbb{F}_{11})$, put $C = \ker H$, and write

$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \quad |\Psi_C\rangle = 11^{-3/2} \sum_{c \in C} |c\rangle.$$

The syndromes whose cosets have minimum X -support three are the 120 nonzero vectors on twelve projective rays forming a nonsingular conic in $\mathbb{P}^2(\mathbb{F}_{11})$. Each such syndrome has exactly one weight-three representative on each of the $\binom{6}{3} = 20$ three-party supports. The monomial automorphism group $C_{10} \times A_5$ is transitive on the 120 syndromes: A_5 is transitive on the twelve rays with point stabilizer C_5 , and C_{10} is transitive on the ten nonzero vectors of each ray. Nevertheless, the defining six-arc is nonconic, so the fixed-party logical symplectic image of every encoder view is the split torus T , not $\mathrm{SL}_2(11)$.

Proof. Lemma 5.3 identifies minimum X -support with coset-leader weight and gives one representative on every three-party support. The Clebsch deep-hole theorem identifies the covering radius as three and the twelve projective maximum-distance syndromes with a nonsingular conic; it also gives the transitive $C_{10} \times A_5$ action [Rud26b, Proposition 6.2, Corollary 6.3, and Proposition 6.4]. Finally, the Clebsch arc is non-GRS and hence nonconic. The last assertion is therefore Theorem 5.2 at $q = 11$. $\square$

Remark 5.5 (Error symmetry without fixed-party duality). The one-per-support statement means that the same extremal translated state can be created by a minimum X -operator on either side of every balanced $3 \mid 3$ cut. The conic belongs to the X -error syndrome plane; it does not put the six parity-check columns on a conic. Thus the extremal X -error shell has icosahedral covariance while the fixed-party transversal group has no Fourier block

exchanging X and Z . This is a fixed-party statement, not an obstruction to every electric–magnetic exchange. In the computed Clebsch row of Corollary C.1, the factor C_{10} scales the ten nonzero syndrome vectors on each ray and disappears after projectivization. The remaining A_5 is the projective monomial quotient, whereas S_5 is the larger realized party image. Odd party motion inverts T and supplies the nontrivial class of $N(T)/T$.

Changing the row basis of H changes the displayed conic by a projective coordinate transformation. Moreover, the covering radius here concerns only generalized- X translates of the six-leg AME state; it is not a claim that the associated $[[5, 1, 3]]_{11}$ output code corrects arbitrary weight-three errors. The unlabelled conic also carries neither a chosen coordinate six-set nor the support orientation used by finer Clebsch constructions. Theorem 1.2 supplies the separate exact rigidity statement: every product-unitary intertwiner from this tensor to a stabilizer AME tensor is Clifford factor by factor.

A different MDS uniformity governs operator pushing. Fix an input party and a nonidentity Pauli label. Each of the ten four-sets containing that party carries a unique stabilizer representative with that support; removing the input leaves a weight-three output Pauli. Thus a fixed input Pauli has ten minimum-support pushes. Lemma 5.3 instead fixes an X -syndrome and varies over all twenty three-party supports. Neither count uses the finite group calculation, although the symmetry orbits of the operator pushes can distinguish finer projective classes.

6 Scalar invariants and generic blindness

The operator-valued marginal mechanism retains local Weyl axes. Fixed-copy scalar contractions retain less: on algebraic families they are generically constant and can distinguish parameters only on rank-jump strata. We make that limitation precise here; Appendix A records two explicit separators on exceptional strata.

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a d -dimensional code, let $M_{\sigma}(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \leq i \leq n, 1 \leq a \leq r).$$

Counting its solutions gives the normalized contraction

$$I_{\sigma}(\Psi_C) = q^{dr - \text{rank } M_{\sigma}(G)}. \quad (6.1)$$

Proposition 6.1 (Generic constancy). Fix r and a field k of cardinality q , which is also the local Hilbert-space dimension under the code–state dictionary. Fix an irreducible reduced parameter variety X over k , and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree- (r, r) permutation contraction of the associated equal-phase states has one value at every k -point of some dense open subscheme $U \subseteq X$ defined over k . This asserts no existence of k -points in U : if $U(k) = \emptyset$, the finite-field pointwise conclusion is vacuous. The same holds for party-orbit sums and linear combinations. After evaluating the states as vectors in $(\mathbb{C}^q)^{\otimes n}$, in the stable range $q \geq r$, every scalar LU-invariant of bidegree (r, r) is generically constant in this sense. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.1) depends only on the rank of a matrix whose entries are regular functions over k . A maximal nonzero minor shows that this rank is constant on a dense open subscheme defined over k . There are only finitely many diagrams at fixed r , so their generic-rank opens have a common dense intersection because X is irreducible. This proves the geometric generic-rank assertion; the explicit caveat in the statement is necessary when passing to finite-field

rational points. Over the complex vector space $(\mathbb{C}^q)^{\otimes n}$, applied pointwise to each code state, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree- (r, r) invariants; when the local Hilbert-space dimension q is at least r , there are no additional dimension relations among these diagrams [CS06, Section 2]. $\square$

Thus no fixed copy number can recover a nonconstant pencil coordinate on the generic locus. A scalar witness can still detect a special rank-jump stratum, as the four-copy example in Appendix A does. In contrast, the four-party reduced-operator family (2.3) retains the Weyl axes, and the imported rigidity theorem forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

7 Verification and scope

The all-length multiplier-line theorem and the exact transversal-group dichotomy are proved in the text. They use no finite census. The imported stabilizer-AME rigidity and phase-correction results are proved in the companion paper [Rud26a]; they are not counted as independent results of this paper.

The six-point applications have a separate exact-arithmetic evidence layer. The paper-local supplement contains the generators, compact JSON certificates, and a deterministic standard-library verifier for the pencil quotient, holonomy reconstruction, Clebsch and H_3 marginal calculations, the four-copy separator at $q = 13$, transport divisor, and twelve party-extension rows. The command

```
python3 supplement/verify.py --replay
```

checks the manifest hashes and regenerates every certificate in memory. The supplement records each finite domain, termination condition, independent replay, and negative scope statement. These computations support the displayed finite applications; they are not premises of Theorem 1.1.

The Lean development is written in Lean 4 with toolchain `v4.32.0-rc1` and a pinned Mathlib revision, in the shared source namespace `RelativeConicArcs.AMELU`. It has two semantic roots: an import-only gate whose transitive closure is this paper’s checked surface, and an audit module that imports only that gate and prints the axiom dependencies of every declaration cited here.

```
gate: RelativeConicArcs.Gates.MDSCSSTransversalGeometry
audit: RelativeConicArcs.Gates.MDSCSSTransversalGeometryAxioms
```

A content-addressed record of that closure, covering its modules, imports, terminal names, axiom facts, exporter digest, and toolchain identity, accompanies the Lean source, and the supplement’s crosswalk names the declarations behind each group of statements below.

Checking by kernel reduction, with no appeal to compiled evaluation, covers the MDS–CSS and stabilizer dictionaries, the diagonal-multiplier line and nullity test, the pencil quotient and the extension-field scalar algebra, syndrome geometry, the contraction and transport cycle algebra, and the abstract consequences of a supplied party-extension complement. The exact carrier, the admitted-pencil classification, the marginal and four-copy separators, the fixed-party logical-phase kernel, and the transport determinant, rank, and orbit-geometry bridges are stated as interfaces. Each derives its conclusion from a structure whose fields assert the geometric propagation, orbit-recognition, or finite-certificate facts that the text or the supplement establishes, so importing the gate proves none of those inputs. Three six-party graph cardinalities are discharged by exhaustive native evaluation, which exposes declaration-local implementation axioms of the pinned toolchain; the rank-four multiplicity identity and

the marginal-moment separator are proved from the star count and inherit its evaluation axiom. Every other cited declaration depends on no more than propositional extensionality, choice, and quotient soundness. The odd-field Weil lift, the Heisenberg nonsplitting argument, the complete translation fiber, and the twelve concrete complements retain the manuscript or certificate boundaries stated above.

The local-unitary classification by z is restricted to odd prime fields. Over extension fields, Frobenius is already an additional local Clifford and the full additive-Clifford orbit problem is not claimed. The exact group theorem is fixed-party and transversal: it does not classify party-moving enlargements outside the twelve certificate-checked rows, non-product physical implementations, nonstabilizer AME states, or unrelated quantum-code families.

8 Conclusion

Diagonal isoduality turns the transversal-group problem into a one-dimensional linear test. MDS distance forces every nonzero code-to-dual diagonal multiplier to be invertible and forces all such multipliers onto one projective line. That line either constructs the elementary unipotents generating $\mathrm{SL}_2(q)$, or its absence leaves only the split torus. Stabilizer AME rigidity then proves that these Clifford constructions exhaust every product-unitary logical implementation.

For six coordinates, the same dichotomy becomes the conic boundary of the six-arc model. The pencil quotient, Clebsch syndrome geometry, and computed party extensions show how the all-length criterion organizes finer finite geometry without depending on those calculations. The generic blindness of fixed-copy scalar contractions marks the remaining methodological boundary: operator-valued frame transport retains information that bounded scalar invariants erase.

Two exact questions remain. Over extension fields, the full additive Clifford group can contain nonsemilinear sectors not detected by the prime-field statement. For party-moving symmetries, the nonabelian factor set gives an exact splitting criterion, but a conceptual classification beyond the twelve computed rows is open.

A Explicit scalar certificates

This appendix records two explicit scalar LU separators. They are not used in the uniform rigidity, atlas, or logical-group results; they show what finite-copy scalar contractions can still detect on exceptional strata despite the generic blindness proved in Section 6.

A.1 A marginal-moment separator

For a four-party set T , let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.3),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\mathrm{rank}(L_C(T)+L_C(U)+L_C(V))}. \quad (\text{A.1})$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12} . Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r} , where r is the $\mathbb{F}_q$ -rank in (A.1).

Index a four-set by its omitted pair, an edge of K_6 . The rank in (A.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X -lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z -lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*). \quad (\text{A.2})$$

where b counts perfect matchings concurrent in both configurations.

Lemma A.1 (Concurrent matchings on a conic). Let S be six distinct points on a nonsingular conic over a finite field k of odd characteristic other than 5. At most six perfect matchings of S have concurrent joining chords.

Proof. Identify the conic with $\mathbb{P}^1$. Projection from a point off the conic pairs the points in every fiber and hence induces a projective involution; conversely, the chords joining the three two-element orbits of a projective involution meet at its center. Thus the concurrent matchings are exactly the fixed-point-free involutions in

$$H = \text{Stab}_{\text{PGL}_2(k)}(S).$$

The correspondence is injective because an element of $\text{PGL}_2(k)$ is determined by its action on three points. Moreover, H acts freely on ordered triples of distinct points of S , so $|H| \mid 6 \cdot 5 \cdot 4 = 120$.

Dickson's classification, including the p -regular cases [Dic01, §§239–261], and Faber's p -irregular classification [Fab23, Theorem 1.1] now leave the following possibilities. The last column bounds involutions having no fixed point on this six-set:

H	fixed-point-free involutions
cyclic	≤ 1
dihedral	≤ 6
A_4	≤ 3
S_4	≤ 6
A_5 in its faithful degree-six action	0
C_3 or S_3 (char $k = 3$)	≤ 3 .

For the A_5 row, each involution fixes two of the six points. In the dihedral row, if the rotation order exceeded six, every orbit away from the two rotation-fixed points would have size at least that order, so no invariant six-set could exist; for rotation order at most six the two reflection classes together contribute at most six fixed-point-free elements (and the possible central half-turn contributes only when one reflection class has fixed points). In the characteristic-three semi-elementary case, the normal 3-group has order 3, while its cyclic complement embeds in $\text{Aut}(C_3)$, giving only C_3 or S_3 . The subfield groups whose order divides 120 are $\text{PSL}_2(3) \cong A_4$ and $\text{PGL}_2(3) \cong S_4$, already listed. The remaining characteristic-five subfield possibilities are excluded by the hypothesis. Every row is therefore bounded by six. $\square$

Theorem A.2 (Uniform H_3 /GRS separation). Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H_3 (icosahedral) sextuple

$$((0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, put $q = |k|$. Its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k . (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H_3 arc and its Gale dual. The other five determinant norms are -64 and -4 , so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H_3 moment multiset therefore contains 70 copies of q^{-4} .

Lemma A.1 gives at most six concurrent matchings on a conic. The bound is attained whenever $\mathrm{PGL}_2(q)$ contains an octahedral S_4 , by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4} , and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X - 3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H_3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (A.1) is LU-invariant, so the gap proves the claim. $\square$

The five nonconcurrent H_3 matchings form a one-factorization of K_6 . The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H_3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

A.2 The exact dimension-thirteen four-copy witness

Use the permutation contraction I_σ and matching system $M_\sigma(G)$ from (6.1).

Theorem A.3 (Exact four-copy separator at $q = 13$). The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\sigma)_i = \sigma_{\pi^{-1}(i)}$, and sum (6.1) over all party permutations:

$$J_\sigma(\Psi) = \sum_{\pi \in S_6} I_{\pi\sigma}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_\sigma(z = 4) = 720 \cdot 13^{-9}, \quad J_\sigma(z = 12) = 3024 \cdot 13^{-9}. \quad (\text{A.3})$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (A.3) makes no claim that four copies are globally minimal. $\square$

Over $\mathbb{Q}(t)$, the matching matrices of every fixed four-copy diagram have generically constant rank. Consequently, after finite-field specialization on the common generic-rank open, their contraction values are constant and do not recover z generically. The witness (A.3) detects a rank-jump stratum. In contrast, the four-party reduced-operator family (2.3) retains the Weyl axes, and the imported rigidity theorem forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

B Transport calculation for the four-copy witness

We now explain the rank jump behind (A.3). Fix the six copy permutations

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \rightarrow k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices $L_a, R_b, 0 \leq a, b < 4$, and a color- i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0. \quad (\text{B.1})$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.1).

Let $U = k^4/k(1, 1, 1, 1)$, and let ρ_i be the action of σ_i on U . For each p , the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^* . Define Q_p by

$$g_{p(i)} = \sum_{k=0}^2 (Q_p)_{ik} g_{p(3+k)} \quad (0 \leq i < 3). \quad (\text{B.2})$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\tilde{Q}_p = 2(t-1)Q_p$. This scalar is a unit on the admitted locus. For the identity assignment, the polynomial representative is

$$\tilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \quad \begin{aligned} a &= -2(t-1)^2, & b &= -(t^2 - t + 1), \\ c &= -(t^2 - 3t + 1), & d &= -2(t-1). \end{aligned}$$

If $Y_k \in U$ are the three free copy vectors, then (B.1) reduces to the 9-by-9 block transport operator

$$\tilde{T}_p = ((\tilde{Q}_p)_{ik}(\rho_i - \rho_{3+k}))_{i,k=0}^2. \quad (\text{B.3})$$

The dimension identity

$$21 - \text{rank } M_{\pi\sigma}(G(t)) = 9 - \text{rank } \tilde{T}_p \quad (\text{B.4})$$

holds over the field k , where $p = \pi$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the admitted six-arc locus.

Theorem B.1 (Transport divisor). For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (B.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z - 2)(9z - 4) = 0. \quad (\text{B.5})$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \quad d^2 = -2a, \quad A = a(c - b), \quad B = bc.$$

Common copy relabelling, a change of basis in V , and rescaling a coordinate covector act on $\tilde{T}_p$ by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3AB(3B - 2A), \quad 64d^3AB(3B + 2A), \quad 64d^3A(B^2 - 2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences organize the exceptional assignments into double cosets. The axial stabilizers have orders 48 and 16, with intersection of order 8, so its support has size

$$48 \cdot 16/8 = 96.$$

For either signed type the corresponding orders are 24, 48, and 6, giving

$$24 \cdot 48/6 = 192.$$

The two signed types are distinguished by the sign of w , while the axial cycle pattern differs from both; hence their support sets are disjoint. Explicit representatives, stabilizer intersections, and complete support sets are in the certificate. Equation (B.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.2. The identity $4B^2 + A^2 = 4G^2$ gives, respectively,

char k	specialized identities
3	$B^2 - 2A^2 = G^2, \quad 9B^2 - 4A^2 = -A^2,$
5	$9B^2 - 4A^2 = 4G^2,$
7	$9B^2 - 4A^2 = 2(B^2 - 2A^2).$

Thus in characteristic 3 both admitted components disappear; in characteristic 5 the $z = 4/9$ component lies on the excluded GRS boundary; and in characteristic 7 the two components merge. The disjoint 96- and 192-element supports then give the exact rank histogram: 288

assignments have rank 20, and 432 have rank 21. Characteristics 11, 13, 41 introduce only pullback ramification, with no further rank drop. These reductions and the nonzero 8-minors that control them are replayed in the certificate.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.1, cannot be promoted to a generic coordinate.

C Computed party-permutation extensions

This optional appendix records the splitting obstruction for the projective party-permutation extension, together with finite examples. It may be skipped without affecting the fixed-party transversal-group theorem.

The nonabelian factor set

Let Γ be the fixed-party projective product-unitary symmetry group, $\tilde{\Gamma}$ allows party motion, and Π is the realized permutation group, so that $1 \rightarrow \Gamma \rightarrow \tilde{\Gamma} \rightarrow \Pi \rightarrow 1$ is exact.

The extension has a section-free outer action $\Pi \rightarrow \text{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ , and changing the lift changes that automorphism by an inner automorphism, because elements of the kernel act by inner automorphisms. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi, \rho)s(\pi\rho), \quad f_s(\pi, \rho) \in \Gamma.$$

Then $f_s(1, \pi) = f_s(\pi, 1) = 1$, and, with α_π denoting conjugation by $s(\pi)$, associativity of three chosen lifts gives

$$f_s(\pi, \rho)f_s(\pi\rho, \nu) = \alpha_\pi(f_s(\rho, \nu))f_s(\pi, \rho\nu).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$ with $b(1) = 1$, then substituting and commuting no kernel factors gives

$$f_{s'}(\pi, \rho) = b(\pi)\alpha_\pi(b(\rho))f_s(\pi, \rho)b(\pi\rho)^{-1}.$$

The order of the factors is essential because Γ need not be abelian. The factor set is identically one precisely when the section preserves multiplication, so the extension splits exactly when f_s can be changed to the constant identity factor set by such a normalized b . That criterion is what the computed rows below verify.

Computed rows

Corollary C.1 (Computed party-extension splittings). The realized projective party-permutation extension splits for both non-GRS $q = 11$ pencil classes, the $q = 11$ GRS evaluation-set representatives

$$\{0, 1, 2, 3, 4, 5\}, \{0, 1, 2, 3, 4, 6\}, \{0, 1, 2, 3, 5, 6\}, \{0, 1, 2, 3, 5, 9\},$$

the enhanced-symmetry $t = -1$ rows over $\mathbb{F}_{17}$ and $\mathbb{F}_{31}$, and the H_3 representatives $(q, \tau) = (11, 4), (19, 5), (29, 6), (31, 13)$. For every listed non-GRS or H_3 row, party parity acts on the fixed split torus T by inversion. Thus, after choosing an anchor leg and allowing party motion to transport the corresponding encoder view, a realized odd party permutation enlarges the possible local linear carrier exactly from T to

$$N(T) = \langle T, J \rangle, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad J \text{diag}(a, a^{-1})J^{-1} = \text{diag}(a^{-1}, a).$$

Hence $N(T)/T \cong C_2$. The linear extension does not split in odd characteristic, although its projectivization does. If the party permutation fixes the input leg of a chosen encoder, this carrier acts as an actual logical enlargement; otherwise it compares two encoder views of the same six-leg tensor.

Proof. For each row, an exact finite computation reconstructs the fifteen four-support shortened planes, exhausts all 720 party permutations, and checks the propagated local symplectic blocks against the full CSS Lagrangian. After quotienting by the fixed-party group identified in Theorem 5.2, it is enough to test anchor block I on the GRS locus and anchor blocks I, J off it; this discards no candidate. In the order in which the six $q = 11$ rows are listed above, their party images are

$$S_5, \quad S_4, \quad V_4, \quad C_5, \quad S_3, \quad D_{12},$$

where D_{12} has order 12. The $q = 17$ and $q = 31$ enhanced rows have party images S_4 and S_5 , respectively, and each listed H_3 image is S_5 .

The certificate records the complete normalized factor set for each section. It also closes a complement of order $|\Pi|$ whose projection to Π is bijective, and verifies that the associated normalized cochain changes the factor set to the identity. Hence every listed extension splits. Conjugating the fixed torus by the stored lifts gives the sign action; any odd lift therefore induces the nontrivial class of $N(T)/T$. The complement is an involution in the projective party-extension group; its determinant-one anchor block is J , whose square is $-I$. $\square$

At $q = 11$, the $t = 2, z = 1$ Clebsch/ H_3 class has party image S_5 , while the $t = 10, z = 9$ non-GRS class has party image S_4 . Both extensions split, and odd party motion inverts T . After choosing an anchor leg, the possible one-qudit affine carriers across transported encoder views therefore form $\mathbb{F}_q^2 \rtimes N(T)$. This is the logical group of a fixed encoder only for the subgroup fixing its input leg; general party motion compares different encoder views. The GRS fixed-party logical group and this anchored non-GRS carrier have orders 159720 and 2420, respectively.

Arithmetic symmetry jumps do not change the logical phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and retains the full $\mathrm{SL}_2(17)$ fixed-party image; its party image is S_4 , and the computed extension splits. The characteristic-31 A_5 specialization is non-GRS, so its fixed-party image is the split torus; its S_5 party image also splits, and odd motion enlarges the transported anchor-block carrier to $N(T)$.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

References

- [Alb26] Victor V. Albert. Beyond transversality: Structure of Clifford circuits for CSS codes, 2026. arXiv:2608.05688v1.
- [CS06] Benoît Collins and Piotr Śniady. Integration with respect to the Haar measure on unitary, orthogonal and symplectic group. *Communications in Mathematical Physics*, 264(3):773–795, 2006.

- [DB25] Shival Dasu and Simon Burton. A classification of transversal Clifford gates for qubit stabilizer codes, 2025. arXiv:2507.10519.
- [DDM03] Jeroen Dehaene and Bart De Moor. The Clifford group, stabilizer states, and linear and quadratic operations over $\text{GF}(2)$. *Physical Review A*, 68(4):042318, 2003.
- [Dic01] Leonard Eugene Dickson. *Linear Groups: With an Exposition of the Galois Field Theory*. B. G. Teubner, Leipzig, 1901.
- [Fab23] Xander Faber. Finite p -irregular subgroups of $\text{PGL}_2(k)$. *La Matematica*, 2(2):479–522, 2023.
- [Gér76] Paul Gérardin. Three Weil representations associated to finite fields. *Bulletin of the American Mathematical Society*, 82(2):268–270, 1976.
- [HDDM05] Erik Hostens, Jeroen Dehaene, and Bart De Moor. Stabilizer states and Clifford operations for systems of arbitrary dimensions, and modular arithmetic. *Physical Review A*, 71(4):042315, 2005.
- [HMSV08] Benjamin Howard, John Millson, Andrew Snowden, and Ravi Vakil. A description of the outer automorphism of S_6 , and the invariants of six points in projective space. *Journal of Combinatorial Theory, Series A*, 115(7):1296–1303, 2008.
- [HP03] W. Cary Huffman and Vera Pless. *Fundamentals of Error-Correcting Codes*. Cambridge University Press, 2003.
- [KKKS06] Avanti Ketkar, Andreas Klappenecker, Santosh Kumar, and Pradeep Kiran Sarvepalli. Nonbinary stabilizer codes over finite fields. *IEEE Transactions on Information Theory*, 52(11):4892–4914, 2006.
- [Rud26a] Tavis Rudd. Local-unitary rigidity and quantitative rounding for stabilizer AME states. <https://doi.org/10.5281/zenodo.21681856>, 2026. Companion preprint; sources at <https://github.com/tavisrudd/ame-lu>.
- [Rud26b] Tavis Rudd. Reconstructing the Clebsch code and its golden orientation from its deep-hole syndrome locus. Preprint, <https://github.com/tavisrudd/clebsch-rigidity>, 2026.
- [SKW⁺25] Hasan Sayginel, Stergios Koutsoumpas, Mark Webster, Abhishek Rajput, and Dan E. Browne. Fault-tolerant logical Clifford gates from code automorphisms, 2025. arXiv:2409.18175.
- [Tan26a] Ian Tan. Classification of five-qubit absolutely maximally entangled states, 2026. arXiv:2507.02185v4.
- [Tan26b] Ian Tan. Transversal gates of the $((3, 3, 2))$ qutrit code and local symmetries of the absolutely maximally entangled state of four qutrits. *Discover Quantum Science*, 1(1), 2026.
- [TTF25] Theerapat Tansuwannont, Yugo Takada, and Keisuke Fujii. Clifford gates with logical transversality for self-dual CSS codes, 2025. arXiv:2503.19790.